3. Problem Modeling

The problem has the following elements (Fig. 1).

- *Decision variables.* For a prescribed number N of identical turbines with rotor radius R (diameter $D = 2R$), the $n = 2N$ coordinates of the layout $\mathbf{X} = (x_1, y_1, \ldots, x_N, y_N)$.
- *Objective.* The expected farm power is maximized, i.e., the wake loss is minimized, under the Jensen wake model [36, 37] and the power curve of [3] (Sections 3.1–3.3).
- *Wind direction.* The direction θ has the probability density p_θ; it is the direction *towards* which the wind blows, counter-clockwise from east (Fig. 1a). The meteorological directions θ_{met} of Horns Rev 1 and IEA37 (wind *from*, clockwise from north) are converted by $\theta = 270^\circ - \theta_{\text{met}}$.
- *Constraints* (Section 3.4). The turbine distances $\ell_{ij} = \|(x_i - x_j,\, y_i - y_j)\|_2$ are at least $\ell_{\min} = 4D = 8R$ [3, 38], and the farm is a disc of radius $r = 500$, 750 or 1000 m.

The spacing $4D$ is the convention of the Kusiak–Song benchmark and is kept for comparability; it lies at the dense end of practice (the installed Horns Rev 1 layout is a regular grid of about $7D$; Section 8.2), so our results concern a packing-constrained regime (larger spacings: Section 11.3).

The model is the unchanged Kusiak–Song benchmark model, and no modeling novelty is claimed; the contributions are the PSO-VNS algorithm and its controlled evaluation. It rests on the following assumptions, each stated in detail in the subsections below:

A1. All N turbines are identical, and their number is prescribed.
A2. The terrain is flat, and the farm boundary is a circle of radius r (blocks of the Horns Rev 1 and Lillgrund farms: their own boundaries).
A3. Turbines must be at least $\ell_{\min} = 4D$ apart.
A4. In each direction bin, the wind speed follows a Weibull distribution; the wind rose is discretized into N_θ direction bins.
A5. Wakes follow the Jensen top-hat model with a constant thrust coefficient C_T; a turbine is waked if its hub center lies in the wake cone, and the deficits of several wakes are combined by the root-sum-square rule.
A6. The power curve is that of the Kusiak–Song benchmark: linear between cut-in and rated speed, constant above.
A7. Only the wake loss is optimized; the electrical collector-system layout and multi-objective formulations (e.g., AEP against cost or the levelized cost of energy) are not considered.

The protocol (equal budgets, seed pairing, equal-cost controls, feasibility-aware ranking, equivalence tests) does not depend on the model and applies to such formulations as well.

3.1. Wind Model and Direction Discretization

For each direction, the wind speed s has the Weibull density

$$p_s(s; \zeta, \psi) = \frac{\zeta}{\psi}\left(\frac{s}{\psi}\right)^{\zeta-1} \exp\left[-\left(\frac{s}{\psi}\right)^{\zeta}\right]$$

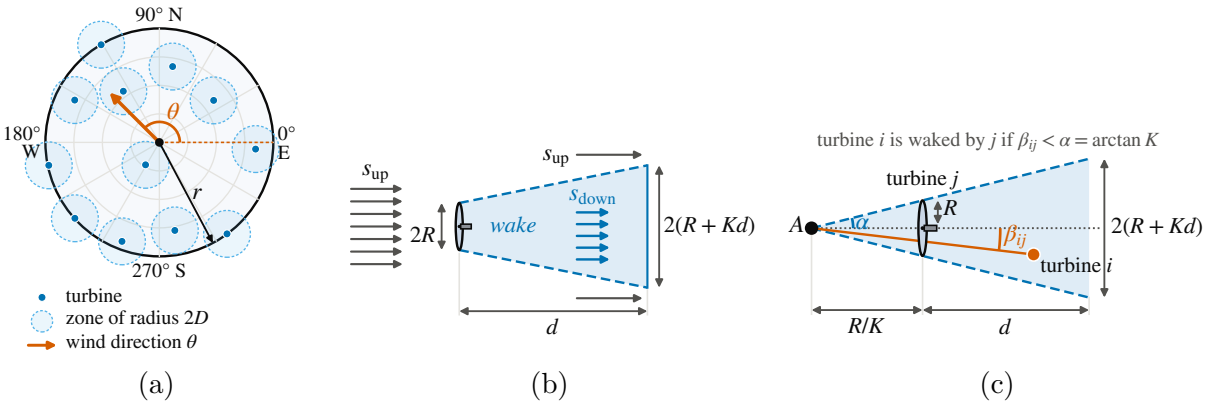


Figure 1: Model geometry. (a) Circular farm of radius r and wind direction θ; the points show, for illustration, a 12-turbine layout ($r = 750$ m); discs of radius $2D$ touch at the spacing $\ell_{\min} = 4D$. (b) Jensen top-hat wake behind a rotor of radius R: width $2(R+Kd)$ at the downstream distance d, uniform speed $s_{\text{down}} < s_{\text{up}}$. (c) Wake half cone of turbine j (K exaggerated): virtual vertex A at R/K upstream of the rotor, half-angle $\alpha = \arctan K$; turbine i is waked by j if $\beta_{ij} < \alpha$.

with shape $\zeta(\theta)$ and scale $\psi(\theta)$. The benchmark rose is piecewise constant: $N_\theta = 24$ direction bins $[15(l-1), 15l)^\circ$, $l = 1, \ldots, 24$, with probability p_l, shape $\zeta = 2$ and scale ψ_l (Table S1). The two wind data sets are:

- *Data Set I*, nearly unidirectional, with the same wind-speed distribution in all directions: $\psi = 13$ m/s in all bins, and 80% of the probability in bins 6 and 7, i.e., wind from 165° to 195°;
- *Data Set II*, a broad rose with direction-dependent wind speeds: scales from 2.6 to 10 m/s and the highest probability in bin 13.

Both are synthetic wind roses of the Kusiak–Song benchmark [3, 38], not measurements at a site; the measured wind climates in this study are the 12-sector Weibull roses of the Horns Rev 1 block (Section 8.2) and of a Lillgrund block (Section 8.3), both optimized in 5° bins, and IEA37 Case Study 1 uses 16 directions at one speed (Section 7.1).

Why the direction discretization matters. The objective evaluates the wake only at the 24 bin centers $\bar{\theta}_l = (15l - 7.5)^\circ$. A top-hat wake is narrow: by the half-cone geometry (Fig. 1c), turbine j wakes a turbine i at the distance $\ell_{ij} \geq R/K$ only for directions within an interval of width $2[\alpha + \arcsin(R/(\ell_{ij}\sqrt{1+K^2}))]$ around the direction from j to i. With $K = 0.075$ and $R = 38.5$ m ($\alpha = 4.29^\circ$, $R/K = 513$ m), this width is below the bin spacing of 15° for $\ell_{ij} > 685$ m and tends to $2\alpha = 8.6^\circ$. If no bin center lies in this interval, the wake of the pair does not enter the objective (with bins of 5° or finer, every such interval contains a center). Optimized layouts exploit these gaps (with 1° sub-bins, the mean wake loss of PSO-VNS rises by about 1 pp); Section 11.5 reports which conclusions survive a 1° rose, for re-evaluated and for directly optimized layouts.

3.2. Wake Effect Model

In the Jensen model (Fig. 1b,c), the wake of turbine j is a half cone with half-angle $\alpha = \arctan K$ (wake-spreading constant K), and its virtual vertex lies R/K upstream of the rotor. With the along-wind distance from the virtual vertex $u_{ij} = (x_i - x_j)\cos\theta + (y_i - y_j)\sin\theta + R/K$ and the lateral offset $v_{ij} = -(x_i - x_j)\sin\theta + (y_i - y_j)\cos\theta$, turbine i lies in the wake if $\beta_{ij} = \cos^{-1}\!\left(u_{ij}/\sqrt{u_{ij}^2 + v_{ij}^2}\right) < \alpha$. Its downstream distance is $d_{ij} = |u_{ij} - R/K|$,

and with the thrust coefficient C_T the velocity deficit, with the free-stream speed s_{up} and the wake speed s_{down}, is

$$\Delta_{ij} = 1 - \frac{s_{\text{down}}}{s_{\text{up}}} = \frac{1 - \sqrt{1 - C_T}}{(1 + K d_{ij}/R)^2}. \tag{1}$$

The numerator is the deficit behind the rotor from actuator-disc momentum theory, and the factor $R^2/(R + K d_{ij})^2$ spreads it over the growing wake (Fig. 1b); with $C_T = 0.8$ and $K = 0.075$, a turbine $\ell_{\min} = 308$ m directly behind another loses 21.6% of the wind speed, one 770 m ($10D$) behind 8.8%. As in [3], this top-hat wake is tested at the hub center only, so the deficit jumps at the cone edge, and it also reaches a turbine less than R/K *upstream* of j near its axis (between A and the rotor in Fig. 1c; non-physical, kept for comparability with the published results). The deficits are combined by the root-sum-square rule,

$$\Delta_i(\theta) = \left(\textstyle\sum_{j \neq i,\ \beta_{ij} < \alpha} \Delta_{ij}^2 \right)^{1/2}. \tag{2}$$

As C_T is constant (which overstates the deficits above rated speed), $\Delta_i(\theta)$ depends only on the layout and the direction, not on the free-stream speed s; the local wind speed in a direction bin, $s_{\text{local}} = [1 - \Delta_i(\theta)]\, s$, is therefore a scaled Weibull variable, with the same shape $\zeta(\theta)$ and the scale $\psi_i(\theta) = \psi(\theta)[1 - \Delta_i(\theta)]$. The Horns Rev 1 model (Section 8.2) instead takes C_T from the V80 thrust curve at the free-stream speed of each 1-m/s speed bin and sums the power over these bins, so it does not use this scaling (Section S1.7). The hub-center test makes the objective jump where a hub crosses a cone edge in an evaluated direction; hence the derivative-free local search of VNS (Section 4.3), and finite-difference gradients handicap MS-SLSQP.

Why the Jensen model. First, the Jensen model with the benchmark settings is needed for comparability with the published Kusiak–Song results [3, 38], and it is cheap enough for the 47,710 optimization runs of this study (Table 4). Second, the research question concerns the optimizers and the protocol rather than the fidelity of the wake model; whether the conclusions carry over to another wake model is an empirical question. Third, we therefore test a Gaussian wake in two ways: the methods are re-optimized under the simplified Bastankhah Gaussian wake of IEA37 Case Study 1 (Section 8.4), and the final benchmark layouts are re-evaluated with the Gaussian wake of [50] (Section 11.2). Control-oriented models, which describe the wake deflection of yawed turbines for wake steering [74, 75], add yaw angles as decision variables and turn the problem into a layout–control co-design, which is outside the scope of this study.

3.3. Power Model and Expected Energy

The power curve of the Kusiak–Song benchmark [3] is

$$f(s) = \begin{cases} 0, & s < s_{\text{cut-in}}, \\ \lambda_1 s + \lambda_0, & s_{\text{cut-in}} \le s \le s_{\text{rated}}, \\ P_{\text{rated}}, & s > s_{\text{rated}}, \end{cases}$$

with $s_{\text{cut-in}} = 3.5$ m/s, $s_{\text{rated}} = 14$ m/s, $P_{\text{rated}} = 1500$ kW, slope $\lambda_1 = 140.86$ and intercept $\lambda_0 = -500$. We use it as published: it is slightly negative just above the cut-in speed (-7.0 kW at 3.5 m/s, zero at 3.55 m/s), jumps from 1472 to 1500 kW at the rated speed and has no cut-out speed. We keep this linearized curve for comparability with

the published Kusiak–Song results; its effect on the energy values is quantified with a cubic curve in Section 11.1, and the Horns Rev 1 block uses the tabulated V80 power curve (Section 8.2). The expected power of turbine i, whose power output $P_i = f(s_{\text{local}})$ depends on its local wind speed,

$$E(P_i) = \int_0^{360} p_\theta(\theta) \int_0^{\infty} f(s)\, p_s\Big(s; \zeta(\theta), \psi_i(\theta)\Big)\, ds\, d\theta, \tag{3}$$

is evaluated, as in [3], by a Riemann sum over direction bins $\theta_0 = 0°, \theta_1, \ldots, \theta_{N_\theta} = 360°$ with probabilities p_l, and over the speed range between $s_{\text{cut-in}}$ and s_{rated}, divided into N_s bins $s_0 = s_{\text{cut-in}}, s_1, \ldots, s_{N_s} = s_{\text{rated}}$ (benchmark: $N_\theta = 24$ bins of 15° and $N_s = 21$ bins of 0.5 m/s). With the bin centers $\bar\theta_l = (\theta_{l-1} + \theta_l)/2$ and the Weibull survival function $W_i(s, \theta) = \exp\{-[s/\psi_i(\theta)]^{\zeta(\theta)}\}$ of the local speed,

$$\begin{aligned} E(P_i) = \sum_{l=1}^{N_\theta} (\theta_l - \theta_{l-1})\, p_l \Bigg\{ & P_{\text{rated}}\, W_i(s_{\text{rated}}, \bar\theta_l) + \lambda_1 \sum_{q=1}^{N_s} \tfrac{s_{q-1}+s_q}{2} \Big[W_i(s_{q-1}, \bar\theta_l) - W_i(s_q, \bar\theta_l) \Big] \\ & + \lambda_0 \Big[W_i(s_{\text{cut-in}}, \bar\theta_l) - W_i(s_{\text{rated}}, \bar\theta_l) \Big] \Bigg\}. \end{aligned} \tag{4}$$

Like the earlier studies, we keep the bin-width factor $(\theta_l - \theta_{l-1})$ in degrees; with 24 bins of 15° and $\sum_l p_l = 1$, the benchmark objective $\sum_i E(P_i)$ is thus 15 times the expected farm power in kW, and the gross (wake-only) annual energy production is AEP [MWh] = (objective/15)×8.76. The wake-free benchmark objective of one turbine, E_{ideal}, is (4) with $\psi_i = \psi$; for Data Set I, $E_{\text{ideal}} = 14{,}045.74$, which corresponds to 936.38 kW (a capacity factor of 62%, far above real sites) or 8,202.7 MWh/yr.

3.4. Optimization Problem and Constraint Handling

With the wake loss $L(\mathbf{X}) = NE_{\text{ideal}} - \sum_{i=1}^{N} E(P_i)$, reported in percent of the wake-free value, $100\, L/(NE_{\text{ideal}})$, the optimization problem reads

$$\begin{aligned} \min_{\mathbf{X} \in [-r,r]^{2N}} \quad & L(\mathbf{X}) \\ \text{subject to} \quad & \ell_{ij} \ge \ell_{\min}, \qquad 1 \le i < j \le N \quad \text{(spacing)}, \\ & x_i^2 + y_i^2 \le r^2, \quad 1 \le i \le N \quad \text{(boundary)}, \end{aligned} \tag{5}$$

which is equivalent to maximizing the expected farm power and hence the AEP: a problem in $2N$ variables ($2N \le 30$ on the benchmark) with a discontinuous objective and $N(N-1)/2$ nonconvex spacing constraints. After every position update, each coordinate is clipped to $[-r, r]$ (box repair; the square, not the circle). The constraints are handled by a penalty on the boundary and spacing violations

$$g_i^{\text{b}} = x_i^2 + y_i^2 - r^2, \qquad g_{ij}^{\text{s}} = \ell_{\min} - \ell_{ij};$$

all optimizers except MS-SLSQP minimize the penalized wake loss

$$\begin{aligned} F_p(\mathbf{X}) = & \Big(NE_{\text{ideal}} - \sum_{i=1}^{N} E(P_i) \Big) + \sum_{i:\, g_i^{\text{b}} > 0} \Big(1 + \mu_{\text{p}} g_i^{\text{b}} \Big)^2 \\ & + \sum_{i<j:\, g_{ij}^{\text{s}} > 0} \Big(1 + \mu_{\text{p}} g_{ij}^{\text{s}} \Big)^2, \qquad \mu_{\text{p}} = 10^{10}, \end{aligned} \tag{6}$$

whose first term is the wake loss $L(\mathbf{X})$ reported in our tables. Three properties of this treatment matter for the comparisons.

- *Feasibility first.* Any violation above about 4.6×10^{-8} (m or m^2) makes the penalty exceed any possible wake loss, so comparing F_p values is, up to smaller violations, a feasibility-first rule (feasible before infeasible, then wake loss or total penalty).
- *Scaling of the violations.* The scaling (g_i^{b} in m^2, g_{ij}^{s} in m) favors infeasible layouts with turbines slightly too close, or with a turbine clipped to the box near a tangent point $(\pm r, y_i)$, where $g_i^{\mathrm{b}} = y_i^2$.
- *Feasibility of the final layouts.* MS-SLSQP minimizes the wake loss under explicit constraints. A final layout counts as feasible if $\ell_{ij} \geq \ell_{\min}$ and $\sqrt{x_i^2 + y_i^2} \leq r$ hold within 10^{-6} m (Section 7), a looser tolerance than the penalty's. With random starts, no optimizer guarantees a feasible final layout; infeasible final layouts are therefore counted, reported and ranked below feasible ones (Section 7.3).

Section 11.4 tests an alternative: Deb's feasibility rules [27] on dimensionless violations, with box clipping or radial projection.

4. Preliminaries: PSO, SSA and Basic VNS

A population method keeps N_p candidate vectors $\mathbf{X}^i \in \mathbb{R}^n$ with components X_m^i and bounds $lb_m = -r$, $ub_m = r$; unless stated otherwise, one iteration costs N_p objective evaluations. This section states the methods exactly as implemented (Algorithms 1–2); Table 2 collects all settings.

4.1. Particle Swarm Optimization

We use the canonical global-best PSO [68] in the inertia-weight form [66] (Algorithm 1), run for T iterations $t = 1, \ldots, T$. Particle i has a position $\mathbf{X}^i$, a velocity $\mathbf{V}^i$ and a personal best $\mathbf{P}^i$, the swarm keeps the global best $\mathbf{G}$, and a particle moves by

$$\mathbf{V}^i \leftarrow w\,\mathbf{V}^i + c_1\boldsymbol{\rho}_1 \circ \left(\mathbf{P}^i - \mathbf{X}^i\right) + c_2\boldsymbol{\rho}_2 \circ \left(\mathbf{G} - \mathbf{X}^i\right), \tag{7}$$

$$\mathbf{X}^i \leftarrow \mathbf{X}^i + \mathbf{V}^i, \tag{8}$$

where $\boldsymbol{\rho}_1, \boldsymbol{\rho}_2$ are vectors of independent $U(0,1)$ numbers, drawn anew for every move, and $\circ$ is the componentwise product. The constriction PSO of Clerc and Kennedy [67] with their standard choice $\varphi_1 = \varphi_2 = 2.05$ (constriction factor $\kappa = 0.7298$) is identical to (7) with

$$w = 0.7298, \qquad c_1 = c_2 = \kappa\varphi_1 = 1.49618. \tag{9}$$

PSO denotes (7) and (8) with these coefficients, fixed a priori from the literature. It is run as implemented: velocities start at zero and are not clamped (no bound $|V_m^i| \leq V_{\max}$), a coordinate clipped to the square (Section 3.4) keeps its velocity, and $\mathbf{P}^i$, $\mathbf{G}$ are updated after each evaluation (asynchronous global best).

Stability under stagnation. Consider one coordinate $x = X_m^i$ with $v = V_m^i$, $p = P_m^i$ and $g = G_m$, and assume (A1) *stagnation*, p and g fixed, and (A2) *no active bound*, so that $v_t = x_t - x_{t-1}$. With $\varphi_{k,t} = c_k\rho_{k,t}$ and $\varphi_t = \varphi_{1,t} + \varphi_{2,t}$, every coordinate then follows the random linear recursion $x_{t+1} = (1 + w - \varphi_t)\,x_t - w\,x_{t-1} + \varphi_{1,t}\,p + \varphi_{2,t}\,g$, whose random coefficient φ_t has mean $\mu = (c_1 + c_2)/2$ and variance $\sigma^2 = (c_1^2 + c_2^2)/12$. The update is

Algorithm 1 PSO for the WFLOP as implemented (complete procedure; $N_p(T+1)$ evaluations of F_p).

Require: site (farm radius r), number of turbines N, wind data and wake model, i.e., the penalized wake loss F_p (6); number of iterations T; random seed
Parameters: swarm size $N_p = 30$; $w = 0.7298$, $c_1 = c_2 = 1.49618$ (9)
Ensure: the global best layout $\mathbf{G}$ and its value $F_p(\mathbf{G})$
Initialization
$n \leftarrow 2N$; draw (seeded) N_p layouts $\mathbf{X}^i \in [-r, r]^n$ and evaluate $F_p(\mathbf{X}^i)$ ▷ N_p evaluations
$\mathbf{V}^i \leftarrow \mathbf{0}$, $\mathbf{P}^i \leftarrow \mathbf{X}^i$ $(i = 1, \ldots, N_p)$; $\mathbf{G} \leftarrow \arg\min_i F_p(\mathbf{P}^i)$
Main loop
for $t = 1, \ldots, T$ **do**
  for $i = 1, \ldots, N_p$ **do**
    Draw $\boldsymbol{\rho}_1, \boldsymbol{\rho}_2 \sim U(0,1)^n$; $\mathbf{V}^i \leftarrow w\,\mathbf{V}^i + c_1\boldsymbol{\rho}_1 \circ (\mathbf{P}^i - \mathbf{X}^i) + c_2\boldsymbol{\rho}_2 \circ (\mathbf{G} - \mathbf{X}^i)$ ▷ (7)
    $\mathbf{X}^i \leftarrow \mathrm{clip}(\mathbf{X}^i + \mathbf{V}^i, -r, r)$ ▷ (8) with box repair; $\mathbf{V}^i$ is kept
    Evaluate $F_p(\mathbf{X}^i)$; if $F_p(\mathbf{X}^i) < F_p(\mathbf{P}^i)$, then $\mathbf{P}^i \leftarrow \mathbf{X}^i$; if $F_p(\mathbf{X}^i) < F_p(\mathbf{G})$, then $\mathbf{G} \leftarrow \mathbf{X}^i$ ▷ asynchronous update
  end for
end for
Termination
Stop after T iterations, i.e., $N_p(T+1)$ evaluations; **return G**

order-1 (*order-2*) *stable* if $\mathrm{E}\, x_t$ (and $\mathrm{E}\, x_t^2$) converges to a limit independent of (x_0, x_1) [69]. Proposition 1 applies these known stagnation results [69, 71] to the implemented update rule; it is not claimed as new theory.

Proposition 1. *Let $c_1, c_2 \geq 0$, $c_1 + c_2 > 0$, and assume (A1)–(A2).*

(a) $\mathrm{E}\, x_t \to x^* = (c_1 p + c_2 g)/(c_1 + c_2)$ *for all initial values if and only if* $|w| < 1$ *and* $0 < c_1 + c_2 < 4(1 + w)$.

(b) The update is order-2 stable if and only if $|w| < 1$ *and*

$$\varrho(1) := 2\mu(1 - w^2) - (1 - w)\mu^2 - (1 + w)\sigma^2 > 0. \tag{10}$$

For $c_1 = c_2$, (10) *is Poli's condition [69]*

$$c_1 + c_2 < \frac{24\,(1 - w^2)}{7 - 5w}; \tag{11}$$

for $c_1 \neq c_2$ with the same sum $c_1 + c_2$, it is stricter.

(c) Under (b), $\mathrm{Var}\, x_t \to V_\infty = (1 + w)\, q/\varrho(1)$ *with* $q = c_1^2 c_2^2 (p - g)^2/[6(c_1 + c_2)^2]$, *for $c_1 = c_2 = c$,* $V_\infty = c(1 + w)(p - g)^2/(4[12(1 - w^2) - c(7 - 5w)])$; $V_\infty = 0$ *if and only if* $c_1 c_2 (p - g) = 0$. *When $c_1, c_2 > 0$, this is equivalent to $p = g$.*

Proof. (a) $e_t = \mathrm{E}\, x_t - x^*$ obeys $e_{t+1} = (1 + w - \mu)\, e_t - w\, e_{t-1}$; both roots of $\lambda^2 - (1 + w - \mu)\lambda + w$ lie strictly inside the unit circle if and only if $|w| < 1$ and $|1 + w - \mu| < 1 + w$ (Schur–Cohn), i.e., $0 < \mu < 2(1 + w)$. (b) With $y_t = x_t - x^*$, $z_t = (\mathrm{E}y_t^2, \mathrm{E}y_t y_{t-1}, \mathrm{E}y_{t-1}^2)^\mathsf{T}$ obeys $z_{t+1} = M z_t + b_t$ with a 3×3 matrix $M(w, \mu, \sigma^2)$ and $b_t \to (q, 0, 0)^\mathsf{T}$, so order-2 stability means a spectral radius of M below one; by Jury's criterion for its characteristic polynomial ϱ, this holds iff $|w| < 1$ and $\varrho(1) > 0$. For $c_1 = c_2 = c$, $\varrho(1) = c\,[12(1 - w^2) - c(7 - 5w)]/6$; for fixed $c_1 + c_2$, σ^2 is smallest at $c_1 = c_2$. (c) V_∞ is the first component of the fixed point of $z \mapsto Mz + (q, 0, 0)^\mathsf{T}$. Since $(1 + w)/\varrho(1) > 0$, the variance vanishes exactly when $q = 0$, including either zero acceleration coefficient. The entries of M and the detailed computations are given in Section S6.1. □

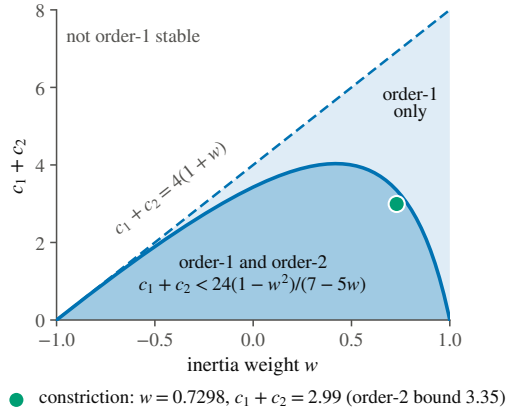


Figure 2: Stability regions of the implemented PSO update under stagnation and without active bounds ($c_1 = c_2$; Proposition 1). Light: order-1 stable only, $c_1 + c_2 < 4(1 + w)$; dark: order-1 and order-2 stable, $c_1 + c_2 < 24(1 - w^2)/(7 - 5w)$. The constriction setting (9) (dot) is order-2 stable.

The order-1 condition involves the *mean* of φ_t. For $c_1 = c_2$, the region also holds under a weaker, non-stagnant assumption [70]. The constriction setting is order-2 stable ($c_1 + c_2 = 2.99 < 3.35$; spectral radius of M 0.94; Fig. 2; Proposition S2): in the limit, a coordinate with $P^i_m \neq G_m$ is sampled with mean $(P^i_m + G_m)/2$ and standard deviation $1.04\,|P^i_m - G_m|$, so the swarm keeps sampling around $\mathbf{G}$ unless $\mathbf{P}^i = \mathbf{G}$. Neither condition says how good $\mathbf{G}$ is, and clipping violates (A2): a clipped coordinate stays at the bound while its kept velocity points outward (Lemma S3). The scale factor $F = 0.5$ of differential evolution (DE; Section 7.1) exceeds Zaharie's critical value (0.14 at CR = 0.9) [76], so its weak results are not due to premature variance loss.

4.2. Salp Swarm Algorithm and Its Laplacian Variant

The SSA was proposed by Mirjalili et al. [8] (pseudocode: Algorithm S1). Its population of N_p salps, sorted by F_p, consists of two halves:

- *Food source.* The best solution found so far, i.e., the candidate with the best fitness (lowest F_p), is the food source $\mathbf{H}$; after every iteration, the best salp of the sorted population, the first leader, replaces $\mathbf{H}$ if it is better.

- *Leaders.* The better half, $i \leq N_p/2$, updates each coordinate around the food source by
$$X^i_m = H_m \pm \xi_1\Big((ub_m - lb_m)\xi_2 + lb_m\Big), \tag{12}$$
with either sign with probability 1/2, $\xi_2 \sim U(0,1)$ and $\xi_1 = 2\exp[-(4t/T)^2]$ decreasing with the iteration t out of T: a leader samples $H_m + U(-\xi_1 r, \xi_1 r)$ in every coordinate, with $\xi_1 \leq 0.037$ for $t \geq T/2$.

- *Followers.* A follower $i > N_p/2$ moves to the midpoint of its position and the *new* position of the preceding salp.

The leaders thus perturb $\mathbf{H}$ uniformly with a shrinking radius, and the followers only average [31]. The SSA has known drawbacks, such as premature convergence and an imbalance between exploration and exploitation [26, 31]. Solanki and Deep [26] addressed them by proposing a new variant, the Laplacian SSA (LX-SSA). LX-SSA gives each follower a second candidate $Y^i_m = X^i_m + \gamma(H_m - X^i_m)$ with one Laplace step [77] for all coordinates,
$$\gamma = \phi - \chi\,\mathrm{sgn}\Big(z - \tfrac{1}{2}\Big)\ln\Big(1 - 2|z - \tfrac{1}{2}|\Big), \quad z \sim U(0,1), \tag{13}$$
with $\phi = 0$ and $\chi = 1$, and keeps the better candidate. As published, it evaluates both candidates of each follower and then re-evaluates the kept ones with the new population,

so an iteration costs $2N_p$ evaluations (SSA: N_p) and LX-SSA runs half as many iterations as SSA (Table 2). Comparing the two tests LX-SSA as published, not the Laplace step alone (for the step itself, see the ablation in Section 9.3); both use the code of the published LX-SSA study [26].

Laplace step. Eq. (13) draws γ by inverse-CDF sampling from a Laplace distribution with location ϕ and scale χ: the sign of $\gamma-\phi$ is that of $z-\frac{1}{2}$, and $|\gamma-\phi| = -\chi \ln(1-2|z-\frac{1}{2}|)$ is exponential with mean χ. With $\phi = 0$, the step is centered on $\mathbf{X}^i$: a move toward $\mathbf{H}$ ($\gamma > 0$) and one away from it ($\gamma < 0$) are equally likely, and, as γ is a scalar, $\mathbf{Y}^i$ lies on the line through $\mathbf{X}^i$ and $\mathbf{H}$ (before box repair). With $\chi = 1$, $\mathrm{E}|\gamma| = 1$, i.e., the mean step equals the distance from $\mathbf{X}^i$ to $\mathbf{H}$, and $P(|\gamma| > 1) = e^{-1} \approx 0.37$, so about 18% of the candidates overshoot $\mathbf{H}$ ($\gamma > 1$); the exponential tails, heavier than Gaussian ones, give occasional long jumps that can aid exploration. The location $\phi = 0$ is that of the Laplace crossover of Deep and Thakur [77], who fix it at zero, and $\chi = 1$ is the value of the published LX-SSA [26]; we keep both untuned. An ablation designed after the main comparison separates the Laplace step from its cost and varies the distribution of γ and the scale χ (Section 9.3): on twelve cases, the step gives no advantage over the SSA midpoint, and $\chi < 1$ is worse.

4.3. Basic Variable Neighborhood Search

The basic VNS [23, 24], in the continuous form of [73] (Algorithm 2), improves one incumbent $\mathbf{X}$ with $k_{\max} = 5$ ℓ_∞ shell neighborhoods $\mathcal{N}_k(\mathbf{X}) = \{\mathbf{X}' : \delta_{k-1} < \|\mathbf{X}' - \mathbf{X}\|_\infty \leq \delta_k\}$, $\delta_k = 0.1\,k\,r$ ($\delta_0 = 0$), of the whole layout: shaking draws a uniform point of $\mathcal{N}_k(\mathbf{X})$, clipped to the box, and a best-improvement local search improves it. As the top-hat wake makes F_p discontinuous, the local search is a compass search: it evaluates all $2n$ moves $\pm h\,\mathbf{e}_m$ ($\mathbf{e}_m$ the mth unit vector; 60 evaluations for $N = 15$), moves to the best one if it strictly improves, and halves h (from $h_0 = 0.05r$) when none does, until $h < h_{\min} = 10^{-3}r$. A completed descent ends at a point that no move $\pm\bar{h}\,\mathbf{e}_m$, $\bar{h} = h_0/2^5 = 0.0015625\,r$, improves (Proposition S9), a coordinate-wise optimality at one step length, not a local minimum.

5. The Proposed PSO-VNS

This section motivates PSO-VNS, states the algorithm and gives its properties. PSO-VNS combines the PSO of Algorithm 1, in the constriction setting (9), with the basic VNS of Algorithm 2 in two consecutive phases.

5.1. Motivation

Three properties of the continuous WFLOP shape the design.

- *Feasible layouts are hard to reach by sampling.* The feasible region is a small, non-convex part of the bounding box; for the largest benchmark cases, fewer than one uniform layout in 10^6 is feasible (Monte Carlo estimate for $N = 12$ and 15; Proposition 2 and Table S48). A method that refines a single layout therefore needs a good, preferably feasible, start, and a population-based global search that moves whole layouts towards the feasible region is a natural source of such starts. PSO is widely used for the WFLOP [6, 7], has few parameters, and its stability under stagnation is understood [69–71], so that a setting can be checked before it is used (Proposition 1).

Algorithm 2 Basic VNS for the WFLOP as implemented (complete procedure; every call of F_p counts against the budget B).

Require: site (farm radius r), number of turbines N, wind data and wake model, i.e., the penalized wake loss F_p (6); budget B; random seed; an incumbent layout $\mathbf{X}$ (VNS alone: none; Phase 2 of a hybrid: the layout passed by Phase 1)
Parameters: $k_{\max} = 5$, $\delta_k = 0.1\,k\,r$ ($\delta_0 = 0$); $h_0 = 0.05\,r$, $h_{\min} = 10^{-3}r$
Ensure: the best layout $\mathbf{X}^\star$ evaluated
Initialization
$n \leftarrow 2N$; **if** no incumbent is given **then** draw (seeded) $N_p = 30$ layouts in $[-r, r]^n$, evaluate them and let $\mathbf{X}$ be the best ▷ VNS alone
$\mathbf{X} \leftarrow$ LocalSearch($\mathbf{X}$); $k \leftarrow 1$
Main loop
while fewer than B evaluations have been used **do**
  Draw $\mathbf{d} \sim U[-\delta_k, \delta_k]^n$ until $\|\mathbf{d}\|_\infty > \delta_{k-1}$ ▷ shaking in $\mathcal{N}_k(\mathbf{X})$
  $\mathbf{X}' \leftarrow \mathrm{clip}(\mathbf{X} + \mathbf{d}, -r, r)$; evaluate $F_p(\mathbf{X}')$
  $\mathbf{X}'' \leftarrow$ LocalSearch($\mathbf{X}'$)
  if $F_p(\mathbf{X}'') < F_p(\mathbf{X})$ **then** $\mathbf{X} \leftarrow \mathbf{X}''$, $k \leftarrow 1$ ▷ move
  else $k \leftarrow k \bmod k_{\max} + 1$ ▷ next, larger neighborhood
  end if
end while
Termination
Stop as soon as B evaluations have been used (also inside LocalSearch); **return** $\mathbf{X}^\star$, the best layout evaluated

function LocalSearch($\mathbf{y}$) ▷ compass search, best improvement
  $h \leftarrow h_0$
  while $h \geq h_{\min}$ **do**
    Evaluate the $2n$ trial layouts $\mathrm{clip}(\mathbf{y} \pm h\,\mathbf{e}_m, -r, r)$, $m = 1, \ldots, n$ ($\mathbf{e}_m$: mth unit vector)
    if the best trial layout $\mathbf{y}'$ has $F_p(\mathbf{y}') < F_p(\mathbf{y})$ **then** $\mathbf{y} \leftarrow \mathbf{y}'$
    else $h \leftarrow h/2$
    end if
  end while
  return $\mathbf{y}$
end function

- *PSO is a weak finisher.* Under a stable setting and while the best positions stay fixed, the expected position of a particle converges to a weighted average of its personal best and the global best [78, 79], a point that need not be a local optimum [71]; late in a run, refinement thus relies on random sampling around the global best. The last improvements of a layout, however, are small moves of individual turbine coordinates, which a coordinate-wise local search makes systematically.

- *The objective is discontinuous.* With the top-hat wake, F_p jumps where a turbine enters or leaves a wake cone (Section 3), so gradient-based refinement is unreliable and a derivative-free local search is needed. The basic VNS [23, 24], in its continuous form [73], fits this need: its local search, a compass search, compares objective values only; its shaking in nested neighborhoods of the whole layout gives a systematic way to leave a local optimum (at the 6,030-evaluation budget studied here it rarely acts; Section 9.1); and its few parameters can be kept at their defaults.

The need for a new variant. PSO and VNS have been combined before, with VNS

as the local search applied within the PSO loop [10–12] (Section 2.5). Such embedded designs spend local-search evaluations in every iteration, which makes the cost of each component hard to separate, and, to our knowledge, they have not been applied to the WFLOP. PSO-VNS instead runs the two methods in sequence: PSO for a share ω of the budget, then VNS from the best layout found. This design

(i) uses the swarm where it is strong, in reaching the feasible region from random starts, and the local search where the swarm is weak, in refining one layout;
(ii) gives both phases fixed, exactly counted budgets;
(iii) adds a single parameter, ω, to those of PSO and VNS; and
(iv) keeps Phase 1 a plain PSO, so that the contribution of the local-search phase can be measured directly against PSO with the same seeds (Section 5.3).

5.2. Algorithm

PSO-VNS (Algorithm 3) is a new two-phase variant of the PSO–VNS hybrid. *Phase 1* runs the PSO of Algorithm 1 from the seeded initial population for $T_1 = \text{round}((\omega B - N_p)/N_p)$ iterations, that is, $N_p(T_1 + 1)$ evaluations, and passes its global best layout $\mathbf{G}$ to Phase 2, feasible or not; the penalty then guides the local search towards feasibility. *Phase 2* applies the basic VNS (Algorithm 2) to $\mathbf{G}$: a compass-search descent, then, until the budget B is used, shaking in the neighborhood $\mathcal{N}_k$ followed by a descent, accepting an improvement and returning to $k = 1$, or else moving to the next, larger neighborhood. The best layout evaluated is returned.

With $\omega = 0.5$ and $B = 6{,}030$, PSO runs for $T_1 = 100$ iterations (99.5 rounded half to even) and VNS receives 3,000 evaluations (Table 2). None of these values was tuned; Section 9.4 studies ω. At this budget, PSO-VNS is essentially PSO plus one compass-search descent, and the neighborhoods matter only for small N (Section 9.1).

5.3. Properties and Computational Cost

Exact prefix. As the PSO coefficients do not depend on the iteration, Phase 1 reproduces exactly, for a given seed, the first $N_p(T_1 + 1)$ evaluations of a PSO run (Proposition S4); comparing PSO-VNS with PSO thus isolates the effect of Phase 2. This does not hold for SSA-VNS and LX-SSA-VNS, whose ξ_1 depends on t/T_1 (Section 6).

Cost. A run uses exactly B evaluations of F_p, including the initial population, and no gradients. Besides the evaluations, both phases need $\mathcal{O}(n)$ arithmetic per evaluation ($n = 2N$): a PSO iteration updates N_p vectors of length n, and a compass-search step forms $2n$ trial layouts; the memory is $\mathcal{O}(N_p n)$. This overhead is small against an evaluation of the wake model, so the run time is governed by B (Section 10.5).

Constraints. Neither phase enforces the constraints: both minimize the penalized loss F_p (6) over the box $[-r, r]^{2N}$, with clipping, and the feasibility of the returned layout is checked and reported for every run (Section 7).

6. Comparison Hybrids and Random-Sampling Controls

6.1. Two-Phase Template

The other hybrids and the random-sampling controls follow the sequential (relay) template of PSO-VNS (Section 5), so that they differ from it only in Phase 1. For a budget of B evaluations and a split $\omega \in (0, 1)$, Phase 1 starts from the seeded initial population, runs a population method for about ωB evaluations and passes its best layout to Phase 2, the basic VNS (Algorithm 2), which applies the local search to it and then iterates until B evaluations are used. Only Phase 1 differs:

Algorithm 3 PSO-VNS for the WFLOP (complete procedure; every call of F_p counts against the budget B).

Require: site (farm radius r), number of turbines N, wind data and wake model, i.e., the penalized wake loss F_p (6); budget B; random seed
Parameters: population $N_p = 30$; $w = 0.7298$, $c_1 = c_2 = 1.49618$ (9); split $\omega = 0.5$; $k_{\max} = 5$, $\delta_k = 0.1\,k\,r$ ($\delta_0 = 0$); $h_0 = 0.05\,r$, $h_{\min} = 10^{-3}r$
Ensure: the best layout $\mathbf{X}^\star$ evaluated, with its wake loss and feasibility
Initialization
$n \leftarrow 2N$; $T_1 \leftarrow \text{round}((\omega B - N_p)/N_p)$ ▷ $T_1 = 100$ for $B = 6{,}030$
Draw (seeded) N_p layouts $\mathbf{X}^i \in [-r, r]^n$ and evaluate $F_p(\mathbf{X}^i)$ ▷ N_p evaluations
$\mathbf{V}^i \leftarrow \mathbf{0}$, $\mathbf{P}^i \leftarrow \mathbf{X}^i$ $(i = 1, \ldots, N_p)$; $\mathbf{G} \leftarrow \arg\min_i F_p(\mathbf{P}^i)$
Phase 1: PSO (global search towards the feasible region)
for $t = 1, \ldots, T_1$ **do**
  for $i = 1, \ldots, N_p$ **do**
    Draw $\boldsymbol{\rho}_1, \boldsymbol{\rho}_2 \sim U(0,1)^n$; $\mathbf{V}^i \leftarrow w\,\mathbf{V}^i + c_1\boldsymbol{\rho}_1 \circ (\mathbf{P}^i - \mathbf{X}^i) + c_2\boldsymbol{\rho}_2 \circ (\mathbf{G} - \mathbf{X}^i)$
    $\mathbf{X}^i \leftarrow \text{clip}(\mathbf{X}^i + \mathbf{V}^i, -r, r)$ ▷ box repair; $\mathbf{V}^i$ is kept
    Evaluate $F_p(\mathbf{X}^i)$; if $F_p(\mathbf{X}^i) < F_p(\mathbf{P}^i)$, then $\mathbf{P}^i \leftarrow \mathbf{X}^i$; if $F_p(\mathbf{X}^i) < F_p(\mathbf{G})$, then $\mathbf{G} \leftarrow \mathbf{X}^i$
  end for
end for ▷ $N_p(T_1 + 1)$ evaluations used
Phase 2: basic VNS (local refinement of $\mathbf{G}$)
$\mathbf{X} \leftarrow$ LocalSearch($\mathbf{G}$); $k \leftarrow 1$
while fewer than B evaluations have been used **do**
  Draw $\mathbf{d} \sim U[-\delta_k, \delta_k]^n$ until $\|\mathbf{d}\|_\infty > \delta_{k-1}$ ▷ shaking in $\mathcal{N}_k(\mathbf{X})$
  $\mathbf{X}' \leftarrow \text{clip}(\mathbf{X} + \mathbf{d}, -r, r)$; evaluate $F_p(\mathbf{X}')$
  $\mathbf{X}'' \leftarrow$ LocalSearch($\mathbf{X}'$)
  if $F_p(\mathbf{X}'') < F_p(\mathbf{X})$ **then** $\mathbf{X} \leftarrow \mathbf{X}''$, $k \leftarrow 1$ ▷ move
  else $k \leftarrow k \bmod k_{\max} + 1$ ▷ next, larger neighborhood
  end if
end while
Termination
Stop as soon as B evaluations have been used (also inside LocalSearch); **return** $\mathbf{X}^\star$, the best layout evaluated in either phase

function LocalSearch($\mathbf{y}$) ▷ compass search, best improvement
  $h \leftarrow h_0$
  while $h \geq h_{\min}$ **do**
    Evaluate the $2n$ trial layouts $\text{clip}(\mathbf{y} \pm h\,\mathbf{e}_m, -r, r)$, $m = 1, \ldots, n$ ($\mathbf{e}_m$: mth unit vector)
    if the best trial layout $\mathbf{y}'$ has $F_p(\mathbf{y}') < F_p(\mathbf{y})$ **then** $\mathbf{y} \leftarrow \mathbf{y}'$
    else $h \leftarrow h/2$
    end if
  end while
  return $\mathbf{y}$
end function

- PSO in *PSO-VNS*;
- SSA or LX-SSA in *SSA-VNS* and *LX-SSA-VNS* (with ξ_1 compressed to the Phase-1 iterations); and,
- in the controls, round(ωB) layouts: the initial population, then layouts drawn uni-

formly in the farm disc (*RSD-VNS*) or in its bounding square (*RS-VNS*).

VNS alone is the limiting case in which Phase 1 is the initial population. Phase 1 runs for

$$T_1 = \mathrm{round}((\omega B - N_p)/c_{\mathrm{it}})$$

iterations of c_{it} evaluations ($c_{\mathrm{it}} = N_p$ for PSO and SSA, $2N_p$ for LX-SSA).

The budget is not shared between methods: each run of each hybrid and control has its own B evaluations, and within the run it is split between the two phases in the same way ($\omega = 0.5$). The hybrids and controls therefore differ only in how Phase 1 spends its half of the budget, which makes their comparison fair. A swarm phase is useful only if it gives VNS a better start than random sampling with about the same number of evaluations (at $\omega = 0.5$, 3,015 for the controls and 3,030 for the swarms ($N_p(T_1 + 1)$; LX-SSA: $N_p(2T_1 + 1)$)); the component analysis (Section 9) tests this for each Phase-1 choice against RSD-VNS, with RS-VNS as a weak minimum bar, as the following elementary result, an application of basic geometric probability to the implemented sampling, shows.

Proposition 2. *Let the $2N$ coordinates be independent and uniform in $[-r, r]$, and let P_N be the probability that the layout is feasible (all turbines in the disc $\mathcal{D}$ of radius r and all pairwise distances $\geq s$). Then (a) all N turbines lie in $\mathcal{D}$ with probability $(\pi/4)^N$; (b) $P_N \leq (\pi/4)^N \prod_{k=1}^{N-1} \max\{0,\ 1 - \max\{A(s),\, k\,A(s/2)\}/(\pi r^2)\}$, where $A(\rho)$ is the area of the part of $\mathcal{D}$ covered by a disc of radius ρ centered on the boundary of $\mathcal{D}$ (closed form: (S8)); (c) among M such layouts, the expected number of feasible ones is MP_N.*

Proof. (a) The turbines are independent and uniform in the square, each in $\mathcal{D}$ with probability $\pi r^2/(2r)^2$. (b) Given (a), the positions $\mathbf{p}_1, \ldots, \mathbf{p}_N$ are independent and uniform in $\mathcal{D}$. If $\mathbf{p}_1, \ldots, \mathbf{p}_k$ are pairwise at least s apart, $\mathbf{p}_{k+1}$ must avoid $U_k = \mathcal{D} \cap \bigcup_{j \leq k} B(\mathbf{p}_j, s)$. The area of $B(\mathbf{c}, \rho) \cap \mathcal{D}$, $\mathbf{c} \in \mathcal{D}$, is nonincreasing in $\|\mathbf{c}\|$ (its derivative is minus the length of the common chord), hence $\geq A(\rho)$; so $|U_k| \geq A(s)$, and, as the discs $B(\mathbf{p}_j, s/2) \subset B(\mathbf{p}_j, s)$ are disjoint, $|U_k| \geq kA(s/2)$. The chain rule over k gives (b). (c) Linearity of expectation. □

The bound (b) is loose; by Monte Carlo, $P_N < 10^{-6}$ for the largest cases ($N = 12$, 15; $N = 10$: no feasible layout in 10^7), so the 3,015 samples of RS-VNS contain about 10^{-3} or fewer feasible layouts in expectation (Table S48), and RS-VNS hands VNS the least-violating sample. The spacing dominates ($P_N \ll (\pi/4)^N$): 730 RS-VNS runs have no feasible sample, and disc sampling gives one in only 242 of them.

Table 2 lists the settings of all methods, the published defaults [8, 26, 67, 73, 80]; MS-SLSQP (SciPy's SLSQP [81]) minimizes L under the scaled constraints of (5) (Section S1.5). At 30,030 and 120,030 evaluations (Section 10.1), T and T_1 follow the same rules.

(Section 7 and later: next review steps)